

Elements of Project Management



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Detailed Syllabus		
Unit	Description	Duration
1.	Introduction to Project Management Definition, classification of project management in different environments, scope, systems approach, project feasibility, project life cycle, systems development cycle phases.	08
2.	Project Organization Project organization, types of structures, roles of project managers and team, integration in large scale projects.	05
3.	Project Planning and Scheduling Planning fundamentals, project master plan, work breakdown structure and other tools of project planning, work packages, project organization structure and responsibilities, responsibility matrix, use of Gantt charts, critical path method (CPM) and project evaluation and review technique (PERT) for scheduling and resource allocation, introduction to software tools for documentation, scheduling, analysis etc.	12
4.	Cost Estimating and Budgeting Cost estimating process, elements of budgeting, project cost accounting and management information systems, cost schedules and forecasts.	06
5.	Project Risk Management Risk concepts and identification, risk response planning, risk management methods, information monitoring, control process, cost accounting systems for project control, performance analysis.	09
6.	Evaluation and Termination Project evaluation and reporting, closing the contract.	05
	Total	45

Text Books

1. Jack R Meredith, Samuel J Mantel, Shafer *Project Management –A Managerial Approach*, 8th Edition, Wiley Publications, 2017.
2. Erik W Larson, Clifford F Gray, Rohit Joshi, *Project Management the Managerial Process*, 8th Edition, McGraw Hill, 2020.

Reference Books

1. Dennis Lock, *The Essentials of Project and Programme Management*, 4th Edition, Routledge Publications, 2017.
2. Prasanna Chandra, *Projects: Planning, Analysis, Selection, Financing Implementation and Review*, 8th Edition, McGraw Hill Education, 2017.

Additional Reference Book:

J. Nicholas, H. Steyn, *Project Management for Business, Engineering & Technology-Principle & Practice*, 5th Edition , Routledge (Taylor & Francis), 2019

NPTEL online course videos

1. Prof. Ramesh Anbanandam – IIT Roorkee
https://www.youtube.com/playlist?list=PLLy_2iUCG87CBuNhvti0h6W54ZmqrSDMJ
2. Prof. R.N. Sengupta -IIT Kanpur
<https://www.youtube.com/playlist?list=PLBd76GK9sWTwVXm9FIVHOTXXbGY2vZR8z>

Note: Project management *knowledge* draws on nine areas, viz., integration, scope, time, cost, quality, procurement, human resources, communications and risk management.

Unit 1: Introduction to Project Management

Definition, classification of project management in different environments ,scope, systems approach, project feasibility, project life cycle, systems development cycle phases

(08 Hours)

Definition:

- ❖ In order to understand project management, one must begin with the definition of a project
 - **Project may be defined as “A temporary endeavor undertaken to create a unique product, service, or result”**
 - **Project** is defined as temporary but interrelated tasks undertaken to give a unique product or service or result
 - **Project** is collection of linked activities, carried out in an organised manner, with a clearly defined START POINT and END POINT to achieve some specific results desired to satisfy the needs of the organisation at the current time
- ❖ **Project management is define as the application of knowledge, skills, and tools necessary to achieve the project's requirements**
 - As per PMI's *PMBOK*® Guide, The knowledge, skills, and tools are grouped into activities or processes;
 1. Project initiation
 2. Project planning
 3. Project execution
 4. Project monitoring and control
 5. Project closure
- ❖ Project management is dynamic process that utilizes the appropriate resources of the organization in a controlled and structured manner, to achieve some clearly defined objectives identified as needs
- ❖ It is always conducted within a defined set of constraints

Project Management = Project + Management

- ❑ A project can be considered to be any series of activities and tasks that:
 - Have a specific objective, with a focus on the creation of business value, to be completed within certain specifications
 - Have defined start and end dates
 - Have funding limits (if applicable)
 - Consume human and nonhuman resources (i.e., money, people, equipment)
 - Are multifunctional (i.e., cut across several functional lines)
- ❑ Examples of project
 - NASA's International Space Station
 - Mumbai Metro
 - Olympic games
 - Nuclear Plant
 - Ships, Skyscrapers
 - Software for Banks
 - Writing book

Definitions of **MANAGEMENT**:

- Management is a process in which the activities of the organizations are managed in a way that ensures the organizational goals are met within desired standards. It is the basis of every organization
- According to **Frederick Winslow Taylor (1856-1915)**, – “Management is the art of knowing what you want to do and then seeing that you do it in the best and the cheapest way”
- According to **George R. Terry (1909-1979)**, “Management is a distinct process consisting of planning, organizing, actuating and controlling, performed to determine and accomplish the stated objective by the use of human beings and other resources
- Definition of management According to **Harold Koontz (1909-1984)**, – “Management is the art of getting things done through others and with formally organized groups. It is the art of creating an environment in which people can perform and individuals can cooperate on the attainment of group goals”
- **Peter Ferdinand Drucker (1909-2005)** “Management may be defined as the process by means of which the purpose and objectives of a particular human group are determined, clarified, and effectuated. Management is a multipurpose organ that manages a business and manages, and the manager manages workers and work
- **Henri Fayol (1841-1925)** – “Management is to forecast, to plan, to organize, to command, to co-ordinate and control activities of others”
- **Harold Koontz & Heinz Weihrich** – Management is the process of designing and maintaining an environment in which individuals, working together, in groups, efficiently accomplish selected aims

What does Project Management Entail?

- **Planning:** is the most critical and gets the least amount of our time

Beginning with the End in mind-Stephen Covey

- **Organizing:** Orderly fashion
(Contingent/Prerequisites)
- **Controlling:** is critical if we are to use our limited resources wisely
- **Measuring :** To determine if we accomplished the goal or met the target?

Measuring.....

Are we efficient?

Are we productive?

Are we doing a good job?

What is the outcome?

Is it what we wanted to be?

***If you can't plan
it, You can't do it
If you can't
measure it, you can't
manage it***



“You have to
know the past
to understand
the present”
— Carl Sagan

EVOLUTION OF PM

1861-1919

The Gantt Chart developed by Henry Laurence Gantt

1950

The critical path method invented (CPM)

1960

The Program Evolution and Review Technique (PERT) invented

1969

Project Management Institute established (PMI)

1975

PROMPTII Methodology created by Simpack systems

1986

Six Sigma Methodology

1989

PRINCE method derived from PROMTII published by U.K. govt

1995

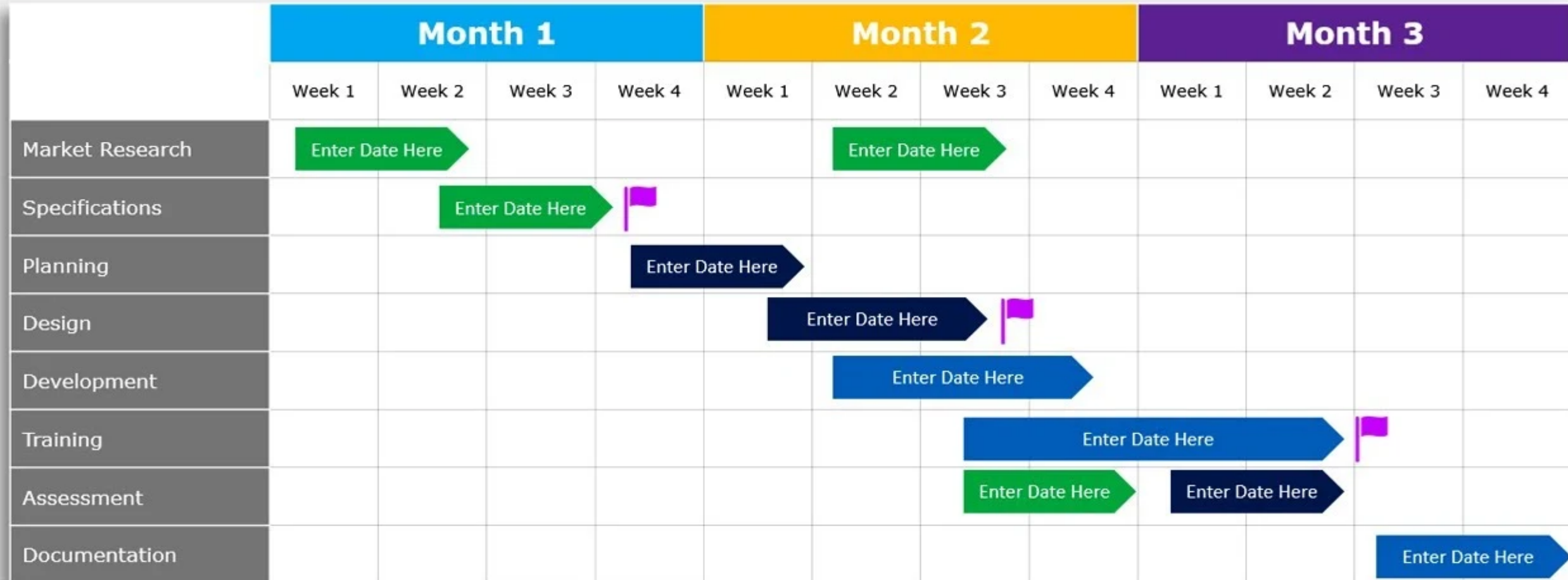
Angle Project Management

2002

Portfolio Management Technique

GANTT CHART

3 Months activity



 Team A

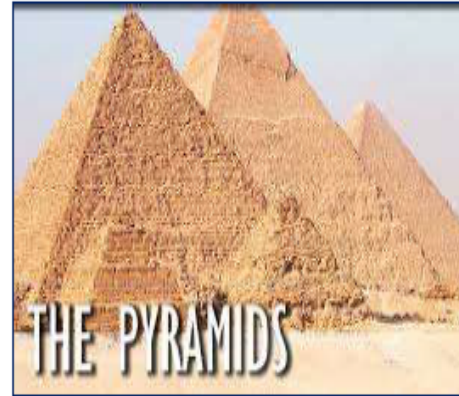
 Team B

 Team C

 Milestones

Evolution of PM

- All of mankind's greatest accomplishments- from building the great pyramid to discovering a cure for polio to putting a man on the moon- began as project.
- Ever since there have been work endeavors that could be defined as "projects", people have been using management tools and techniques.



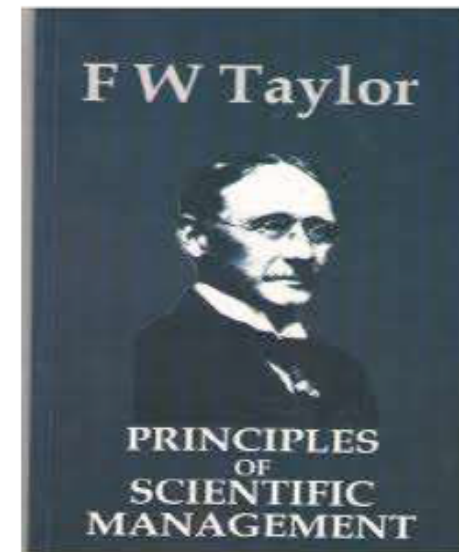
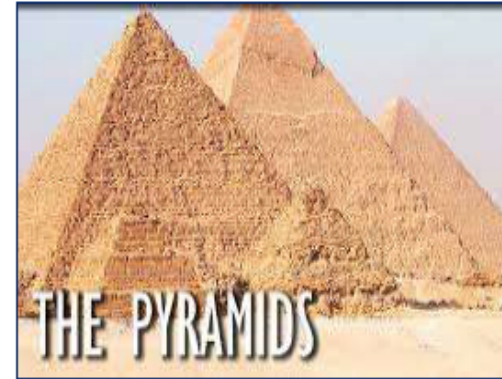
- **The Great Pyramid of Giza (also known as the Pyramid of Khufu)**
 - 481 feet (146.6 meters)
 - Made up of more than 2.3 million massive stones weighing anywhere between 3 and 90 tons each.
 - Survived over 4,500 years of wars, desert storms and other natural calamities
 - According to archaeologists, it took roughly 4,000 primary laborers (quarry workers and masons), supported by 16,000–20,000 secondary workers (ramp builders, tool-makers and suppliers of food, clothing and fuel) to build the massive monument
 - 2 decades, between 2560 B.C. and 2540 B.C.
 - This means the Egyptians laid at least one block—using one Asian elephant—every five minutes

- [illegible]



Project management: History

- Way back in BC 2500...
Pyramids of Egyptians...
(were organized as projects,
but without any management
philosophy.
- 1911 Taylorism/Scientific
Management: Henry L Gantt
invents the Gantt scheme



Project management: History

- 1931: Karol Adamiecki creates the first network diagram, the so called Harmonogram
- 1942-45: The Manhattan project (USA). The project began modestly in 1939, but grew to employ more than 130,000 people and cost nearly US\$2 billion

HARMONOGRAM SZKOLEŃ												
LECZENIE	05.2011	06.2011	07.2011		08.2011	09.2011	10.2011	11.2011		12.2011	01.2012	02.2012
RAJ	I etap				II etap					III etap		
ENG	05.2011	06.2011			08.2011	09.2011	10.2011	11.2011	12.2011		01.2012	02.2012
	I etap				II etap			III etap			IV etap	
KTG		06.2011	07.2011							01.2012	02.2012	
	I etap									II etap		
AKO		06.2011	07.2011	08.2011	09.2011		11.2011	12.2011		01.2012	02.2012	
	I etap			II etap			III etap			IV etap		V etap



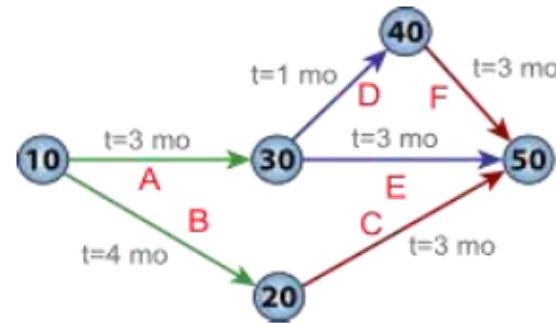
Project management: History

- 1950ies: Operations analysis, RAND Corporation
- 1957: The Sputnik shock, initiating the Polaris project
- The Polaris project: The Polaris missile was nuclear-armed submarine-launched ballistic missile (SLBM) built during the Cold War by **Lockheed Corporation** of California for the United States Navy. Many new project management techniques were introduced during the development of the Polaris missile program, to deal with the inherent system complexity. This includes the use of the **Program Evaluation and Review Technique (PERT)**. This technique replaced the simpler Gantt chart methodology which was largely employed prior to this program.



Project management: History

- 1956-59: CPM (Critical Path Method) is created at DuPont, independent of PERT.

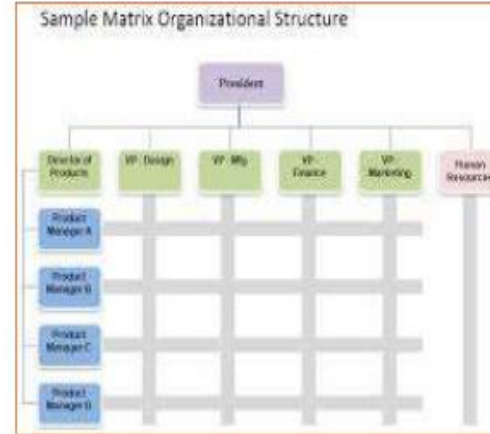


- 1959: The concept of "project manager" is coined in Harvard Business Review.



Project management: History

- 1960ies: Great interest in matrix organizations
- 1967: INTERNET and PMI is established



PMI serves more than five million professionals including over 680,000 members in 217 countries and territories around the world, with 304 chapters and 14,000 volunteers serving local members in over 180 countries.
Revenue: \$343.21 million(Year2021)



Project management: History

- 1980ies: Increasing interest in organizational and project management issues in projects.
The project philosophy spreads to other fields, to smaller activities and to internal activities.
- 1987: **PMBOK** (Project Management Body of Knowledge) presents its first ever certified project management concept by PMI in the form of so called **PMPs** (Project Management Professionals).

Project management: History

- 1990ies: **Management by projects** (the project based company), **Agile project Management**.
- 2000s: Portfolio management techniques.



Who uses Project Management?

- Nearly Everyone to some degree

People plan their Days, their Weeks, their Vacations and their Budgets and keep a simple project management form known as “To Do” list

- Any **Process or Means** used to **track tasks** or efforts towards accomplishing a goal could be considered Project Management

- Project management is a fast-growing field, with more than 25 million new jobs expected to be created globally by 2030
- **Common Industries that Uses Project Management**
 - **Construction**
 - **Manufacturing**
 - **Finance**
 - **Information & Publishing**
 - **Management & Professional Services**
 - **Educational Institutions**
 - **IT-Information Technology**
 - **Healthcare**

Ministry of Finance

- NATIONAL COVID-19 VACCINATION PROGRAMME MEETS ITS GOALS BY OVERCOMING R&D AND LOGISTICAL CHALLENGES, SAYS ECONOMIC SURVEY 2022-23
- UNDER THE PROGRAMME, MORE THAN 220 CRORE COVID VACCINE DOSES ADMINISTRED AS ON 6 JANUARY 2023
- 97 PER CENT OF ELIGIBLE BENEFICIARIES RECEIVE AT LEAST ONE DOSE OF COVID-19 VACCINE
- 90 PER CENT OF ELIGIBLE BENEFICIARIES RECEIVE BOTH THE DOSES
- CO-WIN SYSTEM PROVIDES END-TO-END SOLUTION WITH UTILITIES FOR ENTIRE PUBLIC HEALTH SYSTEM
- CO-WIN PLATFORM PROVIDES REAL-TIME STOCK TRACKING AT THE NATIONAL, STATE, AND DISTRICT LEVELS; HELPS PLUG WASTAGE OF COVID-19 VACCINES

TOTAL VACCINATION DOSE
2,20,68,68,128

Budget

₹35,000 crore (US\$4.2 billion)

India has reported around 43.5 million Covid cases

Health workers involved
92.61 lakhs

❖ Why is Project Management used?

It is necessary to Track or Measure the progress we have achieved towards a Goal we wish to accomplish

We use Project Management to **Aid** us in **Maximizing** and **Optimizing** our resources to accomplish our goals

❖ How much time does Project Management take?

Not much. Probably more time is wasted as a consequence of lack of Project Management tool than is spent to Plan adequately, Organize, Control effectively and Measure appropriately

How long: As long as there are things to do

❖ Consequences of not using Project Management

- DELAY
- COST
- WASTE OF RESOURCES
- QUALITY DISSATISFACTION
- REPUTATION

“The basic principles of project management do not change, but it behooves us from time to time to restate these invariants and our adherence to them”



James Montgomery Beggs
Former NASA Administrator

Classification of Project Management in different environments:

1. Construction Project Management
2. Manufacturing Project Management
3. IT Project Management
4. Biotechnology Project Management
5. Localization Project Management



1. Construction Project Management:

- The construction industry, which focuses on the delivery of things like buildings, roads, and bridges, has developed its own specialized form of project management that it refers to as *construction project management* and in which project managers can become trained and certified

2. Manufacturing Project Management

- Organizations often produce a wide variety of industrial products in the same factory. Introduction of new manufacturing process or product can be taken as a project in an organization.

3. IT Project Management

- The information technology industry has also evolved to develop its own form of project management that is referred to as *IT project management* and which specializes in the delivery of technical assets and services that are required to pass through various lifecycle phases such as planning, design, development, testing, and deployment.

4. Biotechnology Project Management

- *Biotechnology project management* focuses on the projects of biotechnology research and development.

5. Localization Project Management

- *Localization project management* includes many standard project management practices even though many consider this type of management to be a very different discipline.
- It focuses on three important goals: time, quality and budget.
- Successful projects are completed on schedule, within budget, and according to previously agreed quality standards.

Types of Projects in Project Management/Classification of Projects* :

1. Based on ownership

- Public project
- Private project
- Public-Private partnership project

2. Based on investment

- Large scale project
- Medium scale project
- Small scale project

3. Based on sector

- Agricultural projects
- Industrial projects
- Service projects

4. Based on objectives

- Commercial projects
- Social projects

5. Based on nature

- Conventional projects
- Innovative projects
- Technology
- Research
- New product development

6. Based on time

- Long term projects
- Medium term projects
- Short term projects

7. Based on functions

- Marketing projects
- Financial projects
- Human resource projects
- IT & Technology projects
- Production projects
- Strategic projects

8. Based on risk

- High risk projects
- Low risk projects

9. Based on Investment decisions

- Independent Projects
- Mutually exclusive Projects
- Contingent Projects

10. Based on output

- Quantifiable projects
- Non-quantifiable projects

11. Based on Techno-Economic Characteristics

- Factor Intensity Projects
- Causation-Oriented Projects
- Magnitude Oriented Projects

12. Based on Financial Institutions' Classification

- Profit-Oriented Projects
- New Projects
- Expansion Projects or Development projects
- Modernization Projects or Technology Projects
- Diversification Projects

* Ref: Study Material- Project Management,
MBA (Finance), PONDICHERY UNIVERSITY,
DIRECTORATE OF DISTANCE EDUCATION

Scope of Project:

- ❖ Project scope refers to the detailed description of project deliverables, tasks, deadlines, and costs
- ❖ PMBOK defines scope as “ the sum of the products, services, and results to be provided as a project”
- ❖ Project scope describes the reach and scale of the project
- ❖ In simple words scope of project defines what needs to be accomplished within the project and outlines project boundaries, including what is included and what is excluded
- ❖ Project scope varies depending on the number of people involved and the scale of the impact of its outcomes
- ❖ Projects can be big or small depending on the scope
 - The scope document is a critical component of project management as it serves as a roadmap for the project team, stakeholders, and clients
 - Scope document helps project managers and stakeholders manage expectations, allocate resources effectively, and ensure the project stays on track toward its objectives
 - Scope document ensures everyone clearly understands project objectives, requirements, and constraints

- ❑ The essential components of project scope management
 - Scope planning
 - Scope definition
 - Scope verification
 - Scope control

Project management approaches:

- ❖ A project management approach or methodology refers to how a project manager performs a project
- ❖ It's the guideline these professionals use to logically and systematically make decisions about a project
- ❖ An approach influences how project managers lead the rest of the project
- ❖ Types of the project management approaches*
 - Traditional approach
 - Agile approach
 - Waterfall approach
 - Systems approach
 - Critical chain and critical path approach
 - Program Management Approach**

Reference* [Choosing Project Approach Examples \(With How-to Steps\) | Indeed.com Canada](#)

Reference** [5 Most Popular Project Management Approaches That Work \(g2.com\)](#)

Note- For details on systems approach refer: J. Nicholas, H. Steyn, Project Management for Business, Engineering & Technology-Principle & Practice, 3rd edition, Routledge (Taylor & Francis)

❖ **Traditional project management** is the process of planning, executing, and controlling a set of tasks to arrive at the desired goal under budget and within a specified timeframe

❑ **The Agile project management approach** is a highly collaborative methodology that implements feedback as the project progresses to allow flexibility and continuous improvement

❑ There are fewer traditional timelines in this approach, and projects work fluidly

❑ It focuses more on team collaboration than on maintaining a hierarchical leadership structure

❑ This reduces the need for heavy documentation between the design, development, and testing stages

❑ Agile management emphasizes:

1. Working with small batches

2. Working in collaboration with the customer

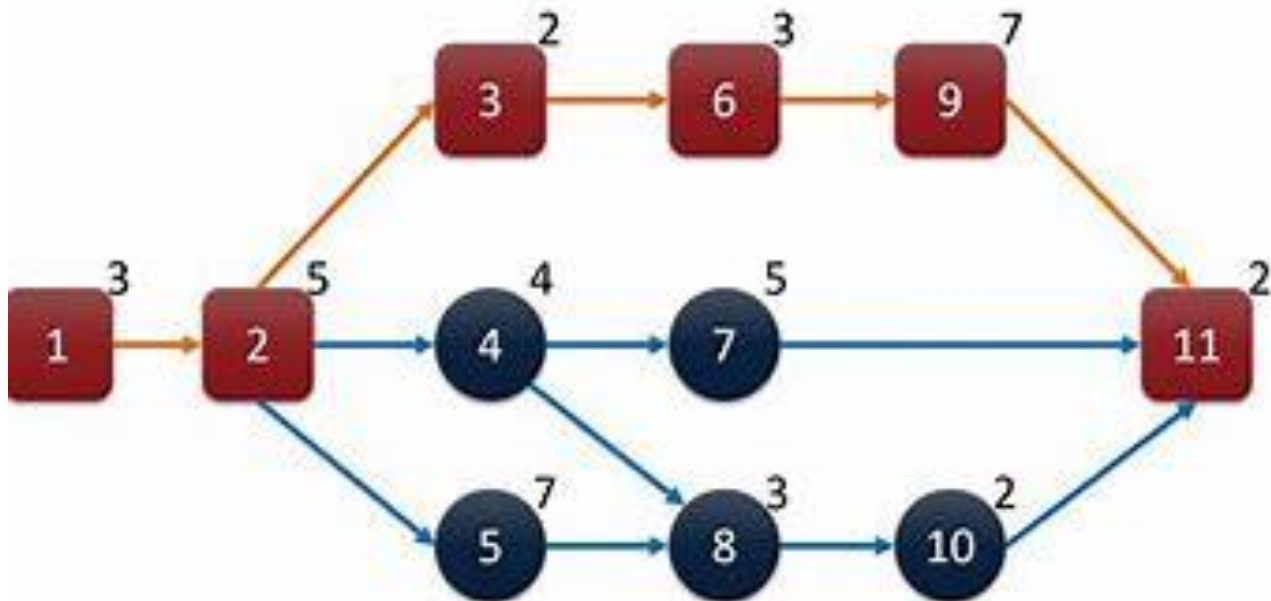
3. Receiving feedback as soon as possible

4. Creating transparency by visualizing processes

- ❖ **The Waterfall approach** is a traditional project management methodology that takes a well-defined project idea to completion through a sequential series of linear steps, tasks, and hand-offs
- ❖ Waterfall approach most often used when the project's goal and its processes are known
- ❖ This approach is used in construction, manufacturing and software development type of industries
- ❖ The Waterfall model was first presented in 1970 by American computer scientist Winston W. Royce in research article titled, "Managing the development of large software systems."
- ❖ This approach may not suit complex projects, as it doesn't allow for modifications and revisions



- ❖ The critical chain project management (CCPM) approach focuses on the resources each task in a project requires, eliminating any constraints that may occur during the project
- ❖ This approach avoids completion delays, unexpected project expenses, and low-quality output
- ❖ The critical path method (CPM) enables project managers to prioritize resources
- ❖ It focuses on completing the most important work while rescheduling lower-priority tasks that delay the project



Critical Path:

- Refers to the most time-consuming tasks and activities.
- Is the longest sequence of tasks that must be completed to execute a project.
- Determines the shortest time possible to complete the project.
- Is identified using the critical path method (CPM) or critical path analysis (CPA)

Critical Chain:

- The critical chain is the sequence of both precedence- and resource-dependent tasks that prevents a project from being completed in a shorter time, given finite resources

What is difference between critical path and critical chain? - Answer from copilot



1. Critical Path:

1. **Focus:** The critical path method primarily manages activities within a project.
2. **Dependencies:** It considers activity dependencies within scheduling.
3. **Resource Assumption:** Assumes that all resources will be readily available when needed.
4. **Objective:** Aims to complete the project in the shortest time possible.
5. **Visualization:** Provides a visual representation of project activities.
6. **Calculation:** Calculates early start, early finish, late start, and late finish dates for all scheduled activities.
7. **Total Float:** Critical path activities have either zero or negative total float.
8. **Use Case:** [Widely used in enterprises following project management best practices¹](#).

2. Critical Chain:

1. **Focus:** Manages the buffer (time reserves) rather than individual activities.
2. **Dependencies:** Considers both activity dependencies and external factors (e.g., resource constraints).
3. **Resource Reality:** Acknowledges resource limitations and constructs a realistic schedule.
4. **Objective:** Aims to avoid time issues and process bottlenecks.
5. **Visualization:** Hones in on essential workstreams required for successful project completion

- ❖ **Program management** is an approach to project management that is useful for managers who have to handle many projects at once
- ❖ Program management is most often used in relation to projects where functional improvement is the main goal, e.g. change management, or industrial engineering

Systems Approach:

A system is “ an organized or complex whole; *an assemblage of parts interacting in a coordinated way towards a common purpose .* ”

- ❖ Systems approach is a way to visualize and analyze physical and conceptual systems, it is an *approach* for *doing* things—a framework for-
 - Conceptualizing problems
 - Solving problems
 - Making decisions
- ❖ System comprises of
 - Sub-System
 - Environment
 - System Boundary
 - Input output Conversion Process

- ❖ The “systems” approach describes a way of dealing with complex problems and opportunities in organizations
- ❖ The systems project management approach enables project managers to track a project at each step to meet the client's needs
- ❖ In this approach, project planning, monitoring, and adapting processes work in a continuous control loop until the project is complete
- ❖ Systems approach assumes that all the project's elements depend on each other
- ❖ Feedback is vital in the systems approach to enable consistent and constant communication

- Organizations are *open* systems, interact with the environment, utilize as inputs ;people, materials, information, and capital, and produce as outputs goods, services, information, and waste byproducts
- Project is a system of people, equipment, materials, and facilities organized and managed to achieve a goal
- The key distinguishing feature of systems thinking is focus on the “ whole organism ” rather than just the parts
- Project managers are concerned with the “ big picture ” —the whole project with its stakeholders and environment

References: 1. Cleland, D. I. (1977). Defining a project management system. *Project Management Quarterly*, 8(4), 37–40

2. J. Nicholas, H. Steyn, *Project Management for Business, Engineering & Technology-Principle & Practice*, 3rd edition, Routledge (Taylor & Francis)

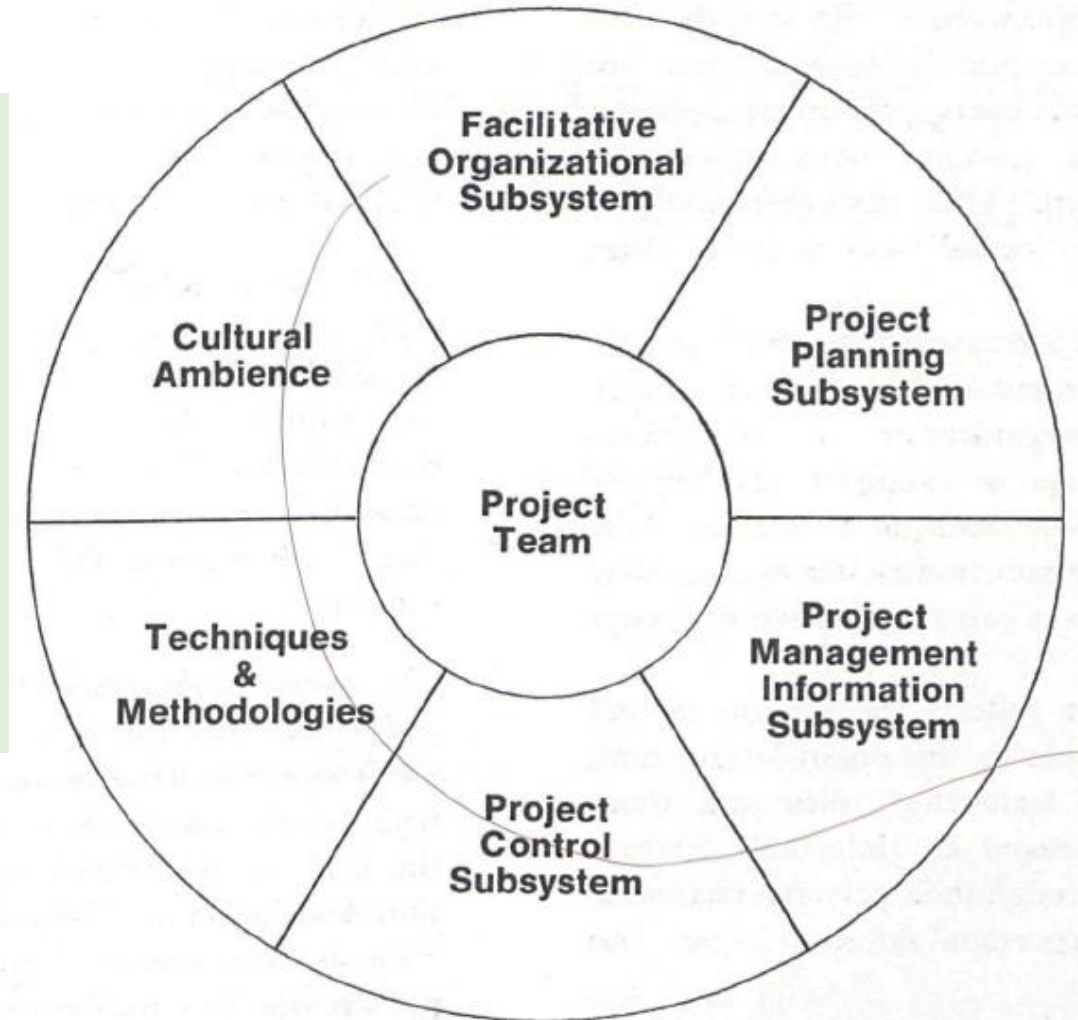
Systems approach to Project Management:

Project management has attained a legitimate place in the evolution of management theory. Concurrent with the emergence of project management has been the “systems approach” to management

**Reference: Cleland, D. I. (1977). Defining a project management system. Project Management Quarterly, 8(4), 37–40*

Figure Depicts-

- A project management system model showing the focal position of the project team and the interacting subsystems
- These subsystems are interdependent
- Their interactions are ultimately reflected in the efficiency and effectiveness with which the project objectives are accomplished
- Success of the project depends on the interactions of these subsystems and how the people perceive the adequacy of the cultural ambience to facilitate effective management of the project



Project Management System*

Note:

- The entity i.e. organization is called a system and its parts are known as sub systems
- David I. Cleland known as Father of Project Management

Additional Points on Systems approach-

The organization as a system receives input, transforms it through a process for output while operating in an environment.

This approach emphasizes that most units within an organization interact with each other and therefore dependent on each other.

E.g. Decision taken by Production department can have impact on other departments like marketing, finance, etc

Additional Points on Systems approach-



Additional Points on Systems approach-

Sub-System:

- A sub system is the operating level of a system. The parts that make up the structure of a system are called sub-systems
- For example, the departments are a sub system of a company. A company in turn can be a sub system of a larger conglomerate

Environment:

- Each system is a part of environment, called “Supra System”.
- It includes all those factors external to a system that affect it and are generally uncontrollable.
- For example, the economic, legal, political and social factors that influence a business system are the environment of that business system.

System Boundary:

- Each system has a boundary, i.e. outer limit of its operations and interaction.
- Open system: A system that dynamically interacts with its environment through a constant procurement of inputs and supply of outputs is an open system.
- Closed System: A closed system does not interact with its environment.

Input-Output Conversion Process:

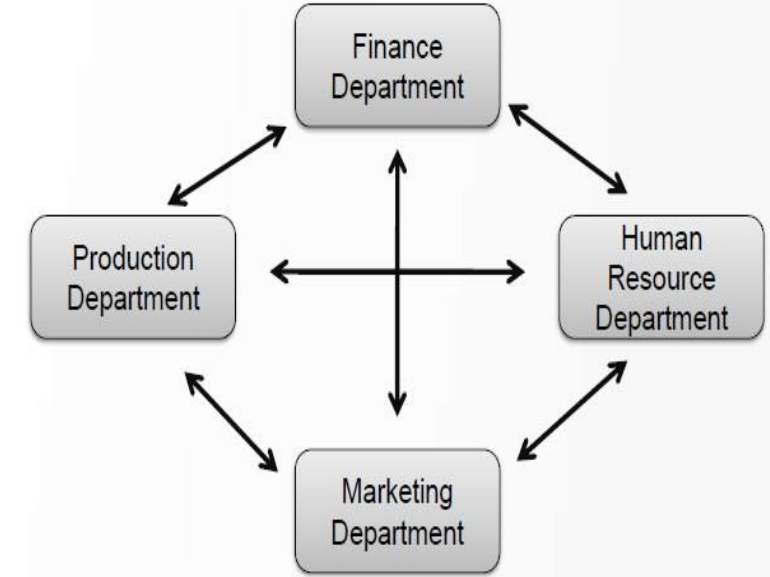
- A system involves input-output conversion process by importing the inputs from environment and transforming them through organizational and managerial processes adopted by sub systems, and exporting the outputs to the external environment.

Feedback: It is a mechanism through which the information about a system's status and performance, in relation to its environment is assessed.

Synergy: It is a state when the output of an entity is in excess of the combined outputs of the constituent units of that entity

Additional Points on Systems approach-

- System approach to management views the organization as a unified, purposeful system composed of interrelated parts
- This approach also gives the manager to see the organization as a whole and as a part of the larger external environment.
- System oriented manager would make decisions only after they have identified impact of these decisions on all other departments and the entire organization.
- They must intertwine their department with the total organization and communicate with all other departments, employees and with each other.



Benefits from Systems Approach:

- ❑ It offers a new framework of organizational knowledge that integrates the elements of structural and management systems of the classical era, psychological and social systems of the neo classical era and the technical system of the management science era
- ❑ It provides an orderly and efficient plan of action.

Weakness from Systems Approach:

- It has been criticized for being less pragmatic and action focused.
- The change in one subsystem will affect other subsystems. But it does not explain the precise direction and magnitude of change.
- It does not offer a unified theory which can be applied to all types of organizations.
- It does not offer specific tools and techniques for the practicing executive

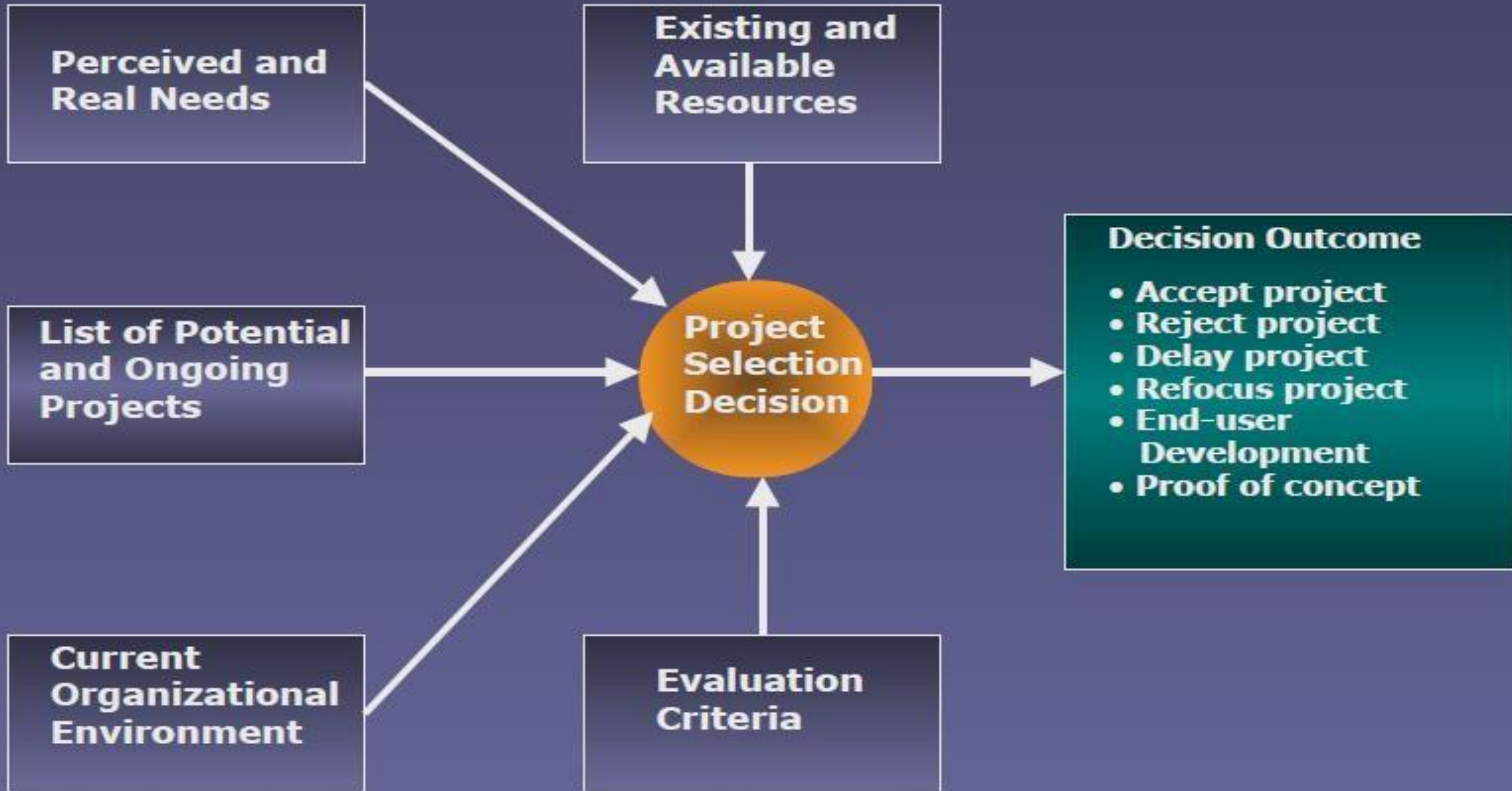
Feasibility:

Feasibility is the process of investigating a need, problem, and solutions in sufficient detail to determine if a solution is economically viable and worth developing.

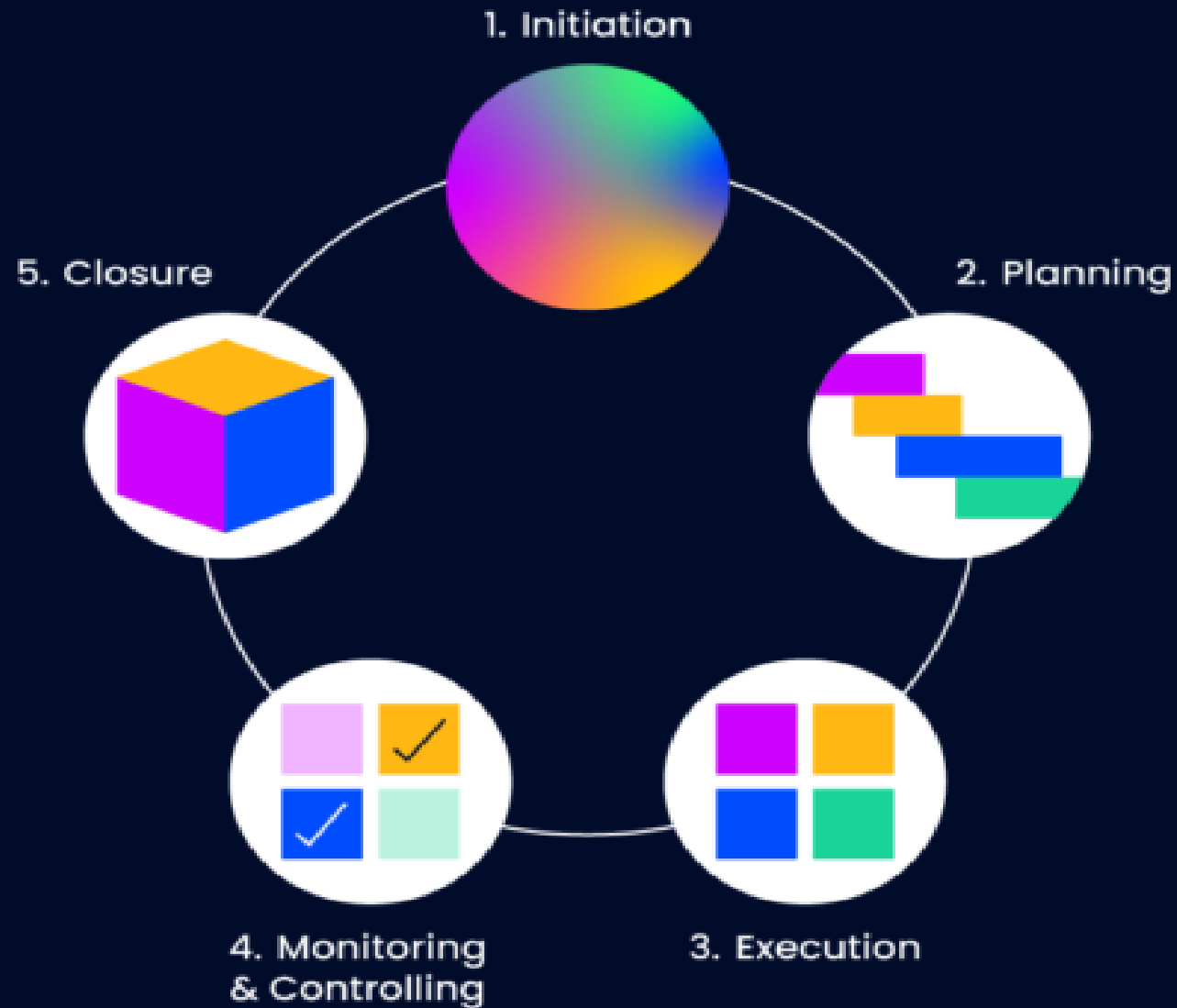
- ❖ Feasibility analysis/study in project management is the systematic evaluation of a proposed project's potential success and viability before committing significant resources to its execution
- ❖ The feasibility study/analysis assess the technical and economic feasibility of a project and serves as a basis for investment and financing decisions
- ❖ For an organization Feasibility is an important planning and organizing procedure to make sure ,project is effective and will work
- ❖ Many projects fail because of unfeasibility and impossibility in their productive measures
- ❖ Types of feasibility studies-
 - Technical feasibility
 - Operational feasibility
 - Economic feasibility

- ❑ A complete feasibility analysis of a project must cover following important study areas*:
 - Technical, including manpower and technological requirements
 - Economic justification, such as the costs and benefits;
 - Administrative/managerial, including external linkages and internal organization;
 - Environmental, including present baseline data and the impact of those data;
 - Social/political, including demographic data and social needs
 - Financial for funding needs and sources

Reference*: Goodman, L.J. (1988). Feasibility Analysis and Appraisal of Projects. In: Project Planning and Management. Springer, Boston, MA. https://doi.org/10.1007/978-1-4684-6587-7_3



Project Life Cycle

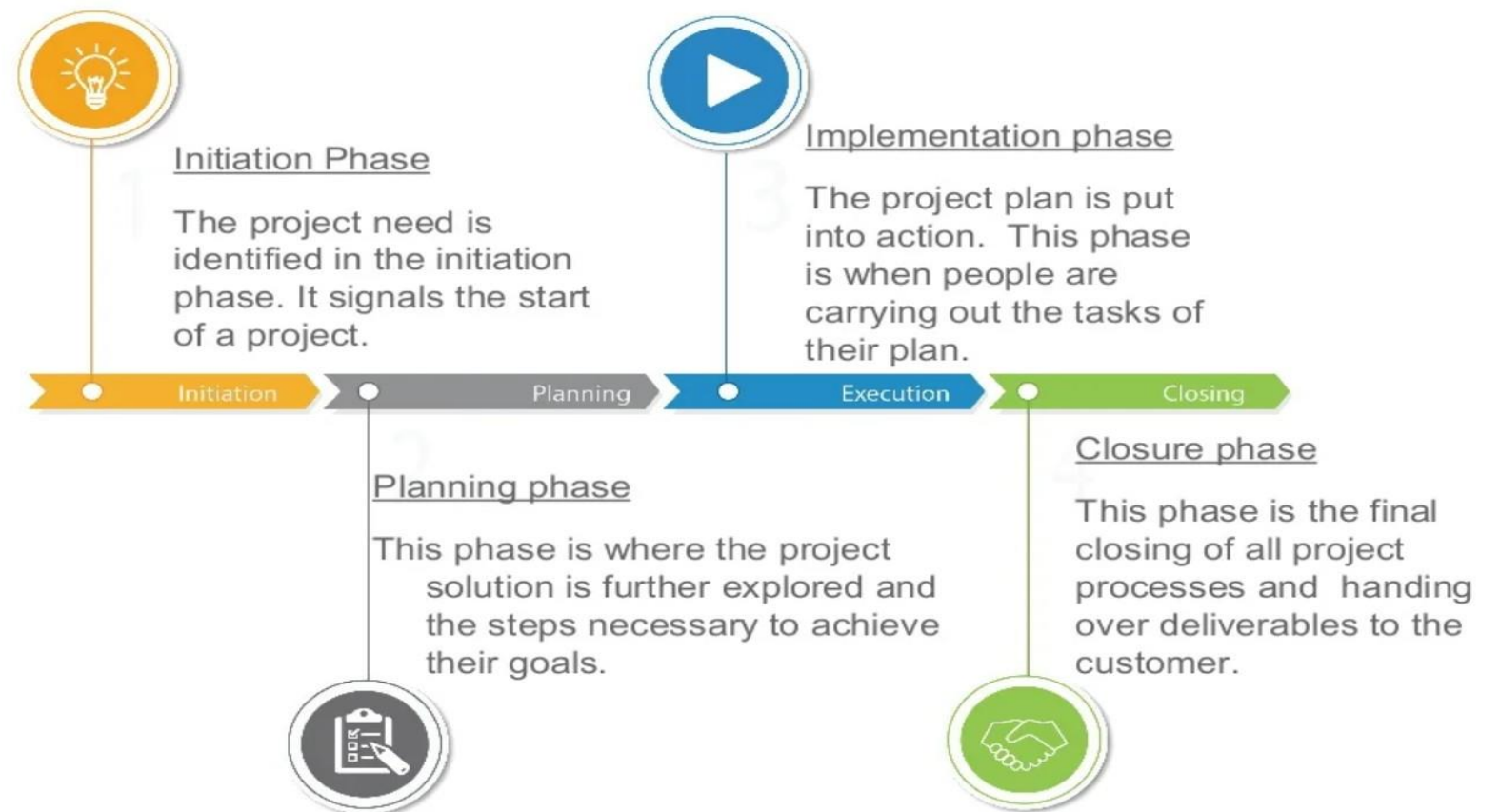


Project Life Cycle:

- ❖ The project life cycle serves to define the beginning and the end of a project (Each project has a starting point and progresses toward a predetermined conclusion)
- ❖ Project life cycle is series of phases of a project from initiation to completion

Project Life Cycle Phases:

- Initiation phase
- Planning phase
- Implementation /Execution phase including Monitoring & control
- Closure phase





Initiation Phase

- The first phase explores the project concept.
- Scope is defined during this phase.
- Feasibility studies are made in order to identify if there is a business need and justification to pursue the project. Project charter is developed for approval.
- This is the phase that the project team is assembled and the project manager is identified.



Planning phase

- This phase further details the scope of the project.
- Tasks and resources are identified and assigned during this phase
- Project manager coordinates the preparation of the schedule and project budget
- Risks are identified ahead to anticipate any project threats
- Quality plan is developed to maintain proper standards throughout project
- Communications plan is created in order to ensure everyone is constantly informed of project status



Implementation phase

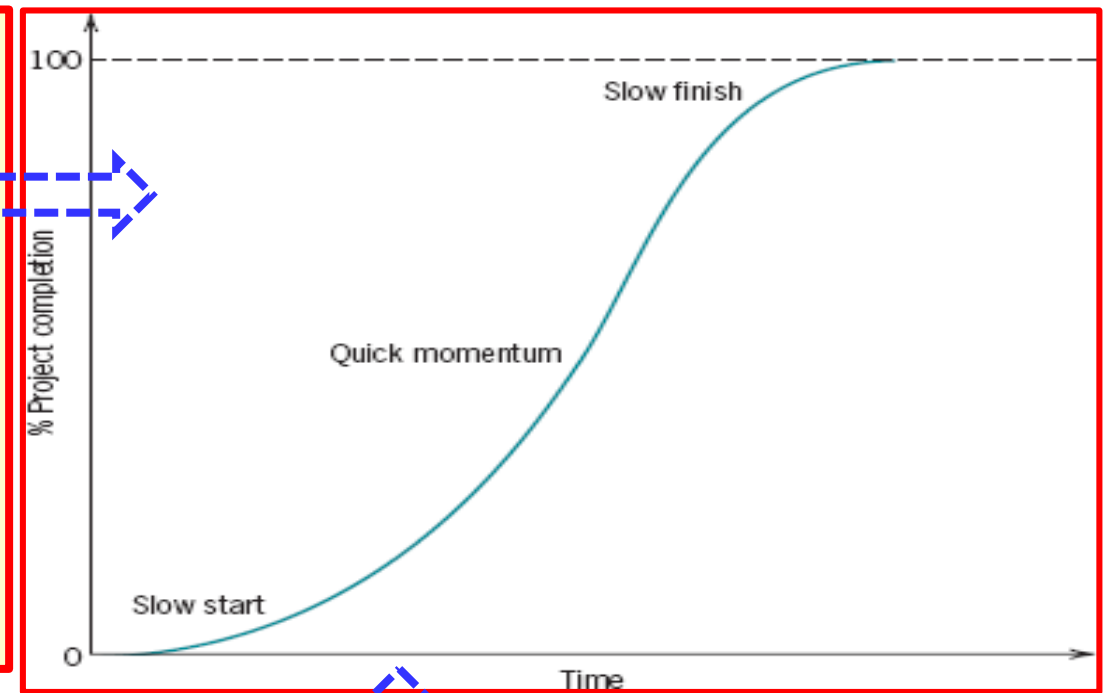
- This phase is where the work outlined in the project plan is performed.
- This phase consumes the most resources and energy.
- Constant and close monitoring of the work should be done to ensure efficiency of the project execution.
- Status reports are important for all stakeholders involved.
- Deliverables are measured against the set metrics to ensure quality is acceptable



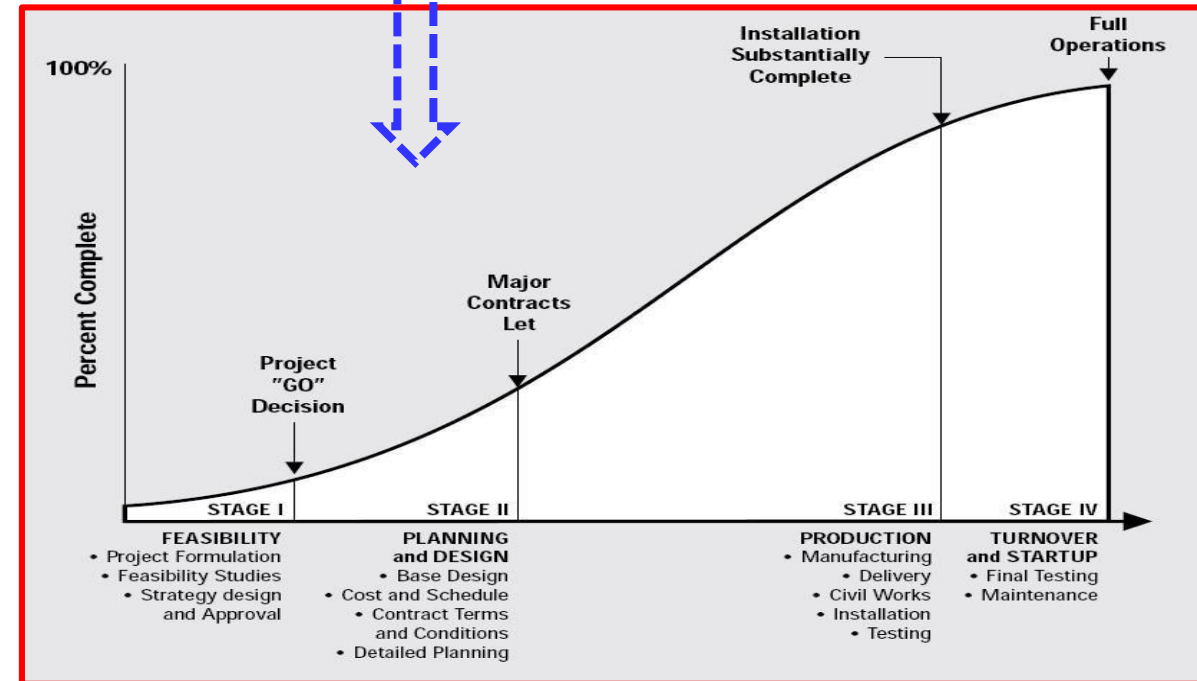
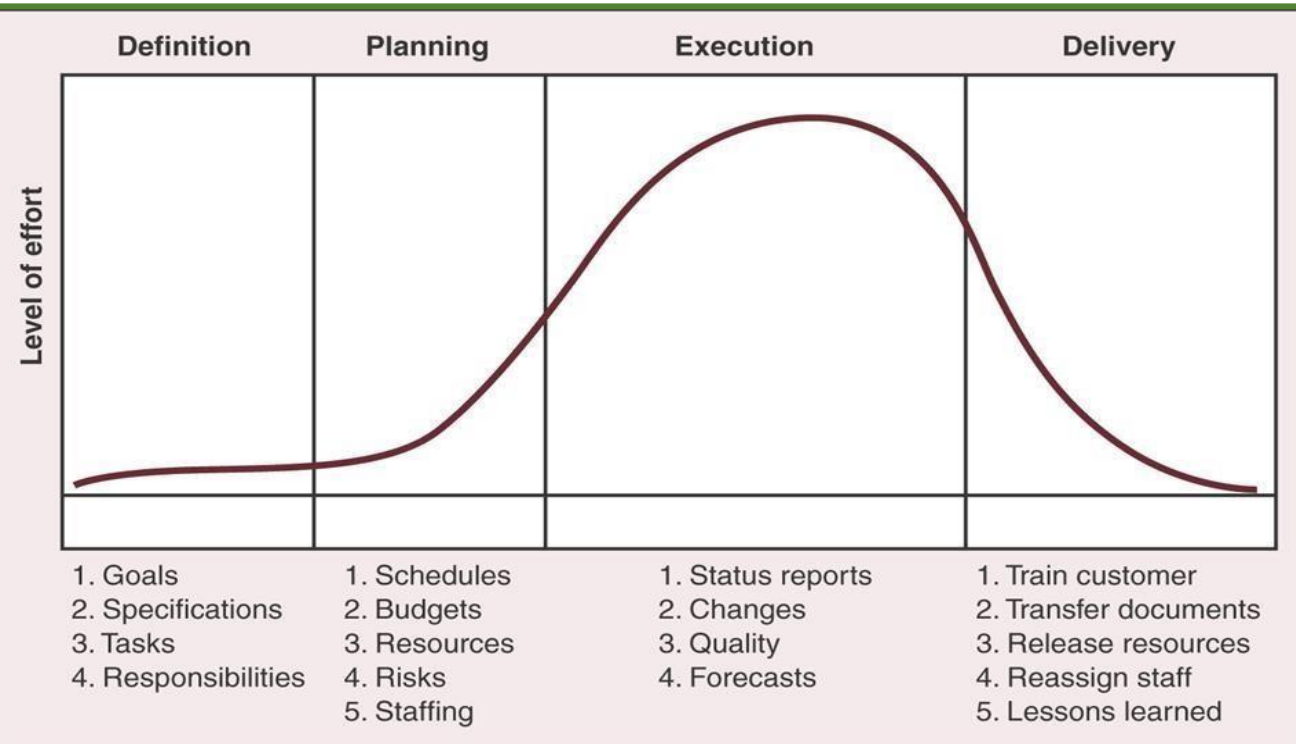
Closure Phase

- This is the last phase of the project life cycle and involves handing over final deliverable to the customer.
- Contracts are properly terminated for equipment, vendors and staff.
- All stakeholders are to be informed of project closure.
- This phase is when the team reviews the overall project and identify lessons learned for future projects.

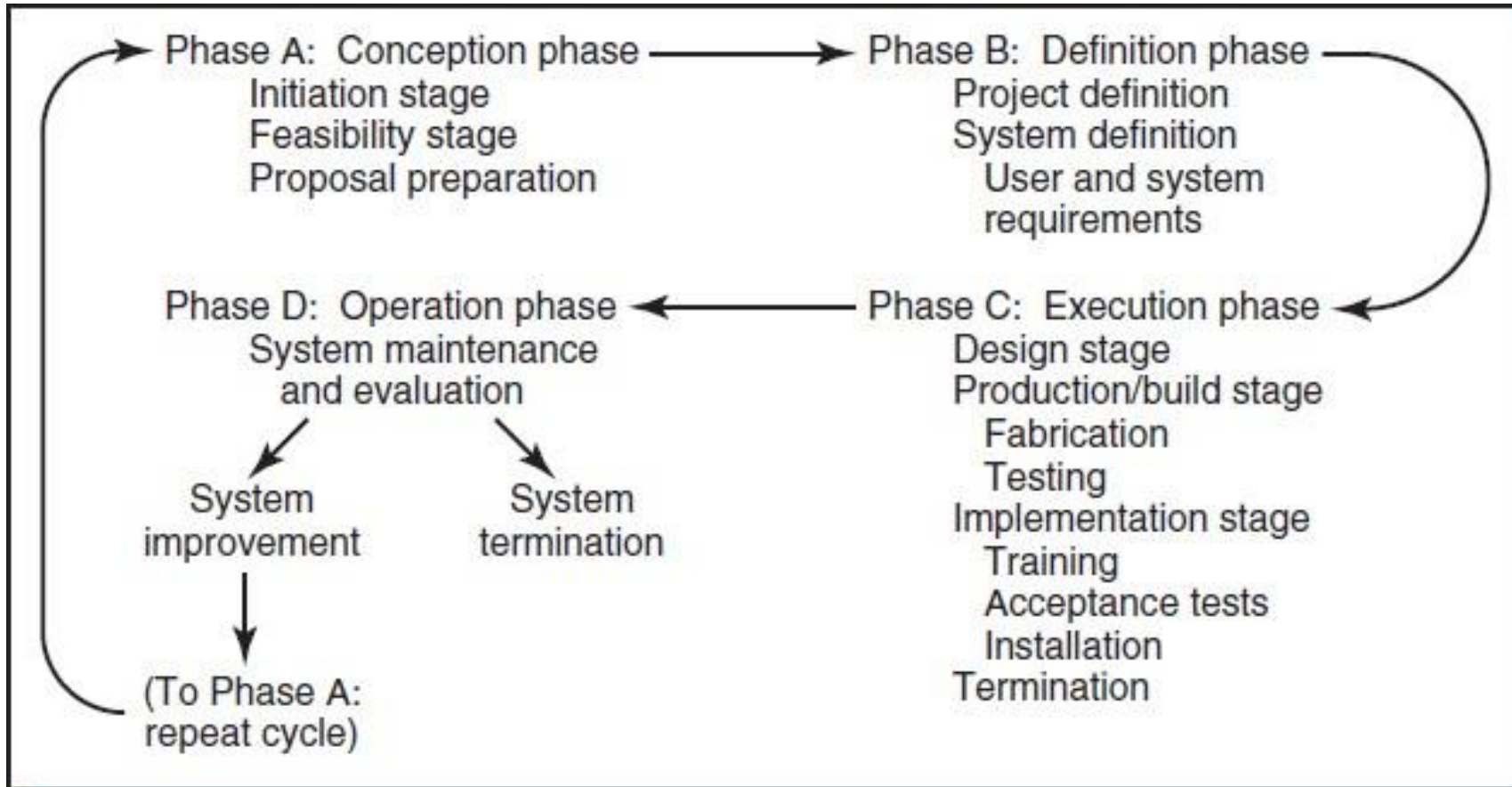
- The project is born - its start-up phase
- manager is selected, the project team and initial resources are assembled and the work program is organized
- Work gets under way and momentum quickly builds
- Progress is made
- Progress continues until the end is in sight
- **Completing the final tasks seems to take an inordinate amount of time**
- number of parts that must come together
- team members “drag their feet” for various reasons and avoid the final steps



The common stretched S- project life cycle



Systems development cycle:



Four-phase model and detailed stages of the systems development cycle.

Project life cycle is Phases A, B, and C

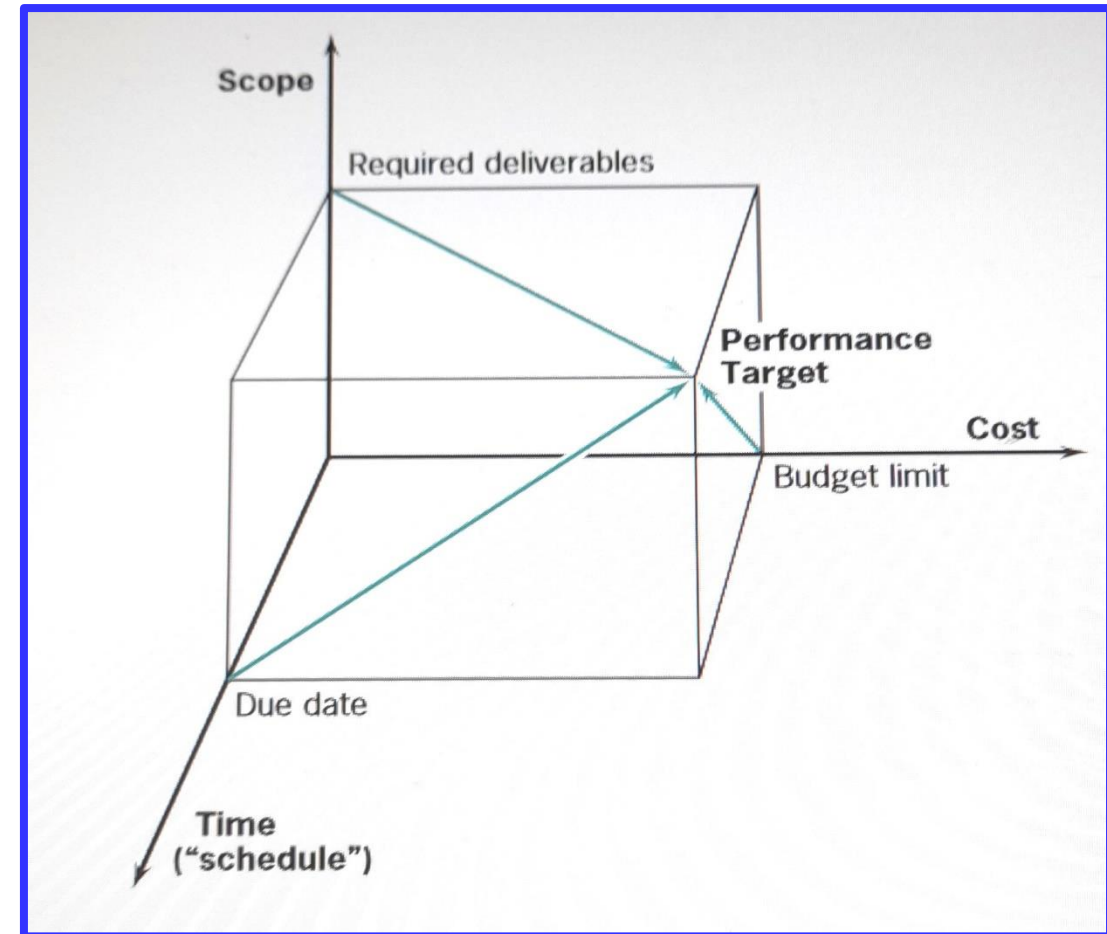
- ❖ The life cycle of a human-made system can be segmented into a logical series of phases and stages
- ❖ System life cycle divided into four phases –called the *systems development cycle*
 1. Conception phase (Phase A)
 2. Definition phase (Phase B)
 3. Execution phase (Phase C)
 4. Operation phase (Phase D)

Triple Constraints of Project Management:

- ❖ The three major constraints of project management-
 - **Time** – Delivering on time
 - **Cost** – Delivering within allotted budget
 - **Scope** – Ensuring adherence to scope of the project



- ❖ In addition to above three constraints, Quality parameters is fourth constraint
- ❖ The goal of every project can be conceptualized in terms of hitting performance target that floats in 3-dimensional space – the dimensions are Time, Cost, and Scope
- ❖ The purpose of project management is to hit the target (to achieve the goal of project)
- ❖ Technological complexity, changing markets and an uncontrollable environment make difficult to hit the target
- ❖ Project Management provides a way to maintain focus on all three major constraints (dimensions)



Need for Effective and Efficient Project Management

- 
1. To control scope of project and manage change
 2. To deliver projects on time and within budget
 3. To ensure the focus of the project team
 4. To collect user requirement from disparate sources
 5. To define the critical path to optimally complete the project
 6. To provide a process for estimating project resources, time and costs
 7. To communicate project progress, risks and changes
 8. To explore project assumptions
 9. To prepare for unexpected project issues
 10. To document the knowledge gained from the project

Characteristics of Project Management**

1. A single person, the project manager, heads the project organization and functions independently of the normal chain of command. The project organization reflects the cross-functional, goal-oriented, temporary nature of the project.
2. The project manager is *the* person who brings together all efforts to meet project objectives.
3. Because each project requires a variety of skills and resources, the actual project work might be performed by people from different functional areas or by outside contractors.
4. The project manager is responsible for integrating people from the different functional areas or contractors who work on the project.
5. The project manager negotiates directly with functional managers who might be responsible for the individual work tasks and personnel within the project.
6. While the project manager focuses on delivering a particular product or service at a certain time and cost, functional managers must maintain an ongoing pool of resources in support of organizational goals. As a result, conflict may arise between project and functional managers over the time and talent to be allotted to a project.
7. A project might have two chains of command, one functional and one project, and people in a project report to both a project manager and a functional manager.
8. Decision making, accountability, outcomes, and rewards are shared between the project team and supporting functional units.
9. Although the project organization is temporary, the functional or subcontracting units from which it is formed are permanent. When a project ends, the project organization is disbanded and people return to their functional or subcontracting units.
10. Project management sets into motion work in numerous supporting functions such as HR, accounting, procurement, and IT.

Potential benefits of effective project management*:

- Clear identification of functional responsibilities to ensure that all activities are accounted for, regardless of personnel turnover
- Minimizing the need for continuous reporting
- Identification of time limits for scheduling
- Identification of a methodology for trade-off analysis
- Measurement of accomplishment against plans
- Early identification of problems so that corrective action may follow
- Improved estimating capability for future planning
- Knowing when objectives cannot be met or will be exceeded

** Ref: J. Nicholas, H. Steyn, Project Management for Business, Engineering & Technology-Principle & Practice, 3rd edition, Routledge (Taylor & Francis)

* Ref: H. Kerzner, Project Management, 12th edition Wiley

Some Important Differences

Project Management	Operation Management
It mainly focuses on management and control of project.	It mainly focuses on management and control of operations.
Project management team have to perform activities like managing projects, streamline works, cost management, etc.	Operation management team have to perform activities like producing products and provide services.
It is a unique task but temporary.	It is a unique task but permanent as it is an on-going process.
Project managers have responsibility of managing starting and closing of project, monitoring different phases of development, etc.	Operations managers have responsibility of delivering services or products for customers in more efficient way.
Focus of project managers changes according to each project requirements and desired results are also unique.	Focus of operations manager does not change, their main focus is to run business operations efficiently and in productive manner.
Project management is quite easy than operation management as responsibilities are less in project management.	Operation management is quite difficult one as responsibilities are more in project management.
Project manager is also responsible for planning, initiating, executing and closing project.	Operation manager is also responsible for making sure that business is running smoothly without any problem to meet goal.
Project management brings change in organization by bringing new products, meet customer demands.	Operation management deliver same product but with new improvements and features as they continuously improve process.

Reference: <https://www.geeksforgeeks.org/difference-between-project-management-and-operation-management/>

Project Management	Program Management
A project is a temporary undertaking task to create a unique product or service, and this is managed by project manager.	In Program management multiple projects are maintained to obtain organisational benefits
The tasks include identifying the sponsor, budget allocation, manage the issues and risks and etc.	The tasks include identifying the program sponsors, Manage program scope and change, Transition solution to multiple operations teams
A project typically exists within one team or within an organisational boundary	A program will often cross multiple organizations, and this has an impact on stakeholders, sponsors, funding sources and even assigning the right program manager.
A project manager is primarily responsible for the scope, cost and timeline.	A program manager can recruit all staff for project management office to manage program scope, costs and overall timeline.
Project managers are used to tracking tasks against a project schedule.	Program managers need to track milestones against a program level schedule
Project management usually revolves around a single site	Program management may involve multiple projects.
Project management is more tactical and it mainly focus on the operational elements of the project	Program management is subject to economic, political, and environmental considerations, especially for publicly-traded organizations.
A project manager isn't responsible for delivering quarterly results.	Program managers are often responsible for delivering results tied to the organization's financial calendar.

Point of Difference	Project Management	Product Management
Goal	Developing plans, implementing strategies, executing the project to completion, and accomplishing business objectives.	Developing a product roadmap, launching a product, collecting feedback, improving customer satisfaction, and enhancing product features.
Scope	Gathering information required to start a project, including the features the product needs to meet its stakeholders' requirements.	Establishing the organization's vision and goals for the product and take all the necessary steps to execute that vision.
Tasks	Planning key performance indicators, coordinating tasks, and gathering resources for projects.	Collecting data, developing roadmaps, launching products, and measuring product success.
Measuring Success	Success is measured by reviewing project specifications, gathering feedback, ensuring budget compliance, meeting schedule needs, and determining quality measurements.	Success is measured by analytical and critical thinking, product launch according to timelines, regularly updation of product roadmap, customer feedback, and retention rate.
Technology Used	Collaboration tools, project tracking software, information-gathering software, scheduling tools, and workflow automation software.	Tracking and analysis tools, roadmapping software, customer survey tools, session replay and heatmap tools, flowcharting tools.

Questions from Previous Exam Question Papers

–MPSTME, Unit-One

Unit 1: Introduction to Project Management

Definition, classification of project management in different environments ,scope, systems approach, project feasibility, project life cycle, systems development cycle phases

(08 Hours)

- a. Describe the key phases of a project's life cycle and their significance [5]
- a. Explain the System Development Life Cycle (SDLC) in the context of Project Management. Provide a detailed overview of the phases involved, their [10]
- Define project management and explain its importance in different environments. [10]

- a. Define project management and explain its importance in various industries. [5]
- a. Explain the systems approach to project management. Provide examples of how [10]
it can be applied to real-world projects.
- a. Outline the phases of a project's life cycle and the significance of each phase. [10]
- a. Discuss the concept of project feasibility. State the key factors to consider when [10]
assessing project feasibility.

- | | | |
|----|---|------|
| a. | Describe the concept 'System Approach' and also explain different elements of system. | [10] |
| a. | Describe the process of how projects get initiated. | [10] |

- | | | |
|----|---|------|
| c. | Explain the concept of Project feasibility. | [5] |
| a. | Describe the concept 'SDLC - System Development Life Cycle' and explain the different phases of SDLC. | [10] |

- Any three questions from A, B, C & any two Sub-Questions from D, E, F
- | | | |
|---|---|-----|
| A | What does Project Management Entail? | |
| A | Define a Project and state advantages of Project Management. | (7) |
| B | Elaborate on system approach in Project Management giving examples. | (7) |
| B | What is SDLC? State and explain stages involved in SDLC. | (7) |